

SUMMARY

I am a data scientist and AI engineer with a scientific approach to problem-solving: formulate the question, build the pipeline, test the hypothesis, and validate the result. I work end to end, from raw data to deployed systems, focusing on reliability, interpretability and measurable impact. My background in analytical research shapes how I design experiments, uncover structure in complex data, and build models that generalize. I combine statistical reasoning with deep-learning methods to create scalable, production-ready solutions for real-world constraints. I question assumptions, iterate rigorously, and prioritize evidence over intuition. My goal is to build data-driven systems that are accurate, stable, and aligned with business needs.

WORK EXPERIENCE

WatchOut • Data Scientist

(Jun 2024) - (Present)

WatchOut is an AI-powered analytics platform specialized in monitoring and analyzing advertising across television, outdoor media, and VOD channels in Iran, delivering real-time insights to brands and marketers.

Tech Stack: Python, OpenCV, PyTorch, Transformers, SQLAlchemy, PostgreSQL, Flask, FFmpeg, Multiprocessing, Multithreading, Vision-Language Models, real-time Monitoring, RAG, Git, Docker, LLM, AI Agents, Prompt Engineering, OOP, MLOps, Fast API

Accomplishments:

- Designed and led the **end-to-end architectures** of Watchout's real-time advertisement intelligence system — from raw video ingestion and frame extraction to AI-driven classification, clustering, and dashboard synchronization — fully developed and maintained independently.
- Built a **multi-process** video deduplication framework that continuously scans and removes repetitive long-form ads from **large-scale databases**, improving dashboard accuracy and preventing duplicate event propagation across pipelines.
- Engineered a real-time ad detection and clustering pipeline capable of identifying unique ads across live TV streams using optimized **Bhattacharyya histogram similarity**, achieving sub-second inference latency and robust frame-level differentiation.
- Fine-tuned** and deployed a custom vision-language model (**VLM**) to power an intelligent sub-clustering system that accurately groups visually similar single-frame ads, drastically reducing misclassification rates in downstream analytics.
- Developed a hierarchical clustering system integrating deep feature extraction (**ResNet-based embeddings** with **PCA** compression) and centroid updates, enabling adaptive categorization of new ad content while preserving data integrity.
- Refactored and extended the system's ad-recognition logic to recover single-frame ads previously missed by older detection algorithms, significantly enhancing overall recall and business coverage.
- Deployed a Flask-based inference **API** service for automated visual similarity evaluation between ad frames, supporting multiple concurrent requests and serving as the backbone for the sub-clustering engine.
- Implemented hourly automated pipelines for all detection, cleaning, and clustering modules, complemented by a continuously running real-time monitoring service; ensured reliability, **scalability**, and self-healing execution across production servers.
- Delivered a **fully autonomous**, fault-tolerant ecosystem that processes thousands of ad instances per hour — integrating database consistency checks, fail-safe rollbacks, and performance-optimized data ingestion routines.

DigiNext, an innovation hub of Digikala Group, fosters emerging tech startups. Within DigiNext, Blue Bee was a data-driven personalization startup focused on developing large-scale recommendation systems to enhance user engagement and tailored experiences

Tech Stack: Python, Dask, SQL, Git, A/B tests

Accomplishments:

- Led the design and deployment of a full-lifecycle personalized product **recommendation engine** built on transaction and user-interaction data, improving user engagement and guiding product discovery across the platform.
- Executed comprehensive exploratory data analysis on the company's online retail dataset to identify **customer segments**, purchasing patterns, seasonal trends and churn risks—driving strategic insights for marketing and product teams.
- Designed, conducted and interpreted **A/B-tests** for website funnel optimizations and promotional campaigns—defining hypotheses, selecting metrics, running statistical tests and translating results into actionable decisions to boost conversion rate and average order value.

CCG-SBU is a computational data-analysis research lab affiliated with Shahid Beheshti University (SBU) in Tehran. Their goal is to study cosmological problems such as cosmic tensions and parameter degeneracies, aiming to produce more precise measurements of cosmological parameters.

Tech Stack: Python, SciPy, Distributed-Computing Tools, ML/deep-learning, Git, Bash, Linux

Accomplishments:

- Designed and implemented a workflow to preprocess large-scale simulation data, extract multivariate features, and apply topological data-analysis techniques (persistent-homology algorithms) to characterize complex structural patterns in the data.
- Built and maintained a modular pipeline (data ingestion → feature generation → Betti-curve/persistence-diagram computation → statistical summarization) that supports automated scanning of parameter grids and redshift bins.
- Transformed persistence-diagrams into machine-learning-ready representations (e.g., persistence images, embeddings) and experimented with classification/regression models to identify distinguishing signals across simulation outputs.
- Developed and trained convolutional-autoencoder based models to mask unwanted foreground contamination in large image/map datasets, enabling extraction of cleaner target signals and improving downstream analysis robustness.
- Owned the full project lifecycle—from problem definition, algorithm development, coding, execution of batch simulation runs, to result interpretation and documentation—independently and end to end.

EDUCATION AND CERTIFICATIONS

- **M.Sc. in Computational Physics** (2021) - (2023)
 - *Shahid Beheshti University, Tehran.*
- **B.Sc. in Physics** (2016) - (2021)
 - *Shahid Beheshti University, Tehran.*